

Grade 8
Unit 13
Lesson 7 Practice

Name _____
Date _____
Period _____

1. How many sides does a regular polygon have if the sum of the interior angles is 3960° ?

2. What is the measure of one interior angle of the regular polygon found in the first question? Round to the nearest tenth if necessary.

Use the following for questions 3–5.

The measure of an interior angle in the regular 20 sided figure can be represented by the expression $(19.3x + 7.6)^\circ$.

3. Find the sum of the interior angles.

4. Find the measure of one interior angle.

5. What is the value of x ?

Use the following for questions 6–8.

The measures of two interior angles of a regular polygon are $(14x - 28.2)^\circ$ and $(27 + 10x)^\circ$.

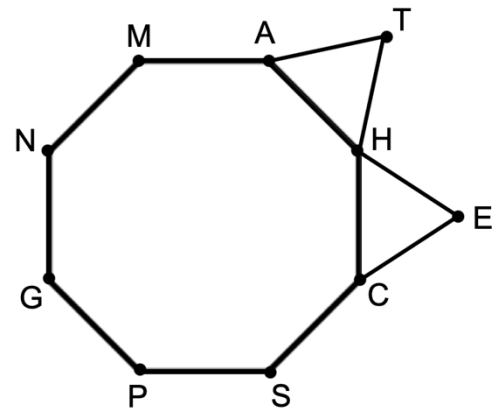
6. Write and solve an equation to find the value of x .

7. What is the measure of each interior angle?

8. How many sides does the regular polygon have?

A bathroom floor consists of tile in the shape of regular octagons and equilateral triangles.

9. Find the sum of $\angle MAH$ and $\angle HAT$.



10. The measure of $\angle MAT$ is $(25x + 15)^\circ$. Determine the value of x .